

# Analog IC Design ASSESSMENT 4

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**Submitted to**  
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# INDEX

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|    |                                                                    |
|----|--------------------------------------------------------------------|
| 1  | Objective and Specifications                                       |
| 2  | Theory and Transfer Functions                                      |
| 3  | Part A – Resistive-Load CS Amplifier                               |
| 4  | Part A – Device Characterization and $g_m/I_D$ Sizing              |
| 5  | Part A – Small-Signal Model and Pole-Zero Analysis                 |
| 6  | Part A – Simulated Gain, $-3$ dB Response and Unity-Gain Frequency |
| 7  | Part A – Gain Margin and Phase Margin                              |
| 8  | Part B – Current-Mirror / Active-Load CS Amplifier                 |
| 9  | Part B – Separate NMOS and PMOS Sizing by W Sweep                  |
| 10 | Part B – Small-Signal Model and Transfer Function                  |
| 11 | Part B – DC, Transient, AC and Pole-Zero Analysis                  |
| 12 | Part B – Simulated Gain, $-3$ dB Response and Unity-Gain Frequency |
| 13 | Part B – Gain Margin and Phase Margin                              |
| 14 | Comparison and Conclusion                                          |

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## Section 1: Objective and Specifications

The experiment involves the design and simulation of two inverting common-source amplifiers.

For Part A, the specified voltage-gain magnitude is 4 V/V, the unity-gain bandwidth is 100 MHz, and the load capacitance is 5 pF.

For Part B, the same gain, unity-gain bandwidth and load capacitance are specified together with a 1 mW power-consumption target.

**Table 1:** Design specifications

| Parameter | Part A: Resistive Load | Part B: Active Load |
|-----------|------------------------|---------------------|
| $V_{DD}$  | 1.8 V                  | 1.8 V               |
| $ A_V $   | 4 V/V                  | 4 V/V               |
| $f_u$     | 100 MHz                | 100 MHz             |
| $C_L$     | 5 pF                   | 5 pF                |
| $P_{DC}$  | –                      | 1 mW                |

## Section 2: Theory and Transfer Functions

### 2.1 Common-Source Amplifier

For a MOS common-source stage operating in saturation, the mid-band voltage gain is approximately

$$A_V \approx -g_m R_{out}. \quad (1)$$

For the resistive-load CS stage,

$$R_{out} \approx R_D \parallel r_o \quad (2)$$

and

$$A_{V,A} \approx -g_m (R_D \parallel r_o). \quad (3)$$

For the active-load stage,

$$R_{out} \approx r_{oN} \parallel r_{oP} \quad (4)$$

and

$$A_{V,B} \approx -g_m (r_{oN} \parallel r_{oP}). \quad (5)$$

The negative sign indicates inversion.

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## 2.2 $g_m/I_D$ Design Method

The required transconductance is obtained from

$$g_m = 2\pi f_u C_L. \quad (6)$$

The drain current is

$$I_D = \frac{g_m}{g_m/I_D}. \quad (7)$$

Using the characterized drain-current density,

$$W = \frac{I_D}{I_D/W}. \quad (8)$$

Thus, the MOS bias / characterization circuit is used to determine the transistor width of the final amplifier.

## 2.3 Transfer Function

The amplifier transfer function is

$$H(s) = \frac{V_{out}(s)}{V_{in}(s)}. \quad (9)$$

A practical amplifier may be represented as

$$H(s) = A_0 \frac{1 - s/\omega_z}{(1 + s/\omega_{p1})(1 + s/\omega_{p2})}. \quad (10)$$

The dominant pole determines the first major high-frequency roll-off, while zeros and higher-order poles determine the later frequency-domain behaviour.

## 2.4 Control-System Definitions

For a feedback-loop transfer function  $L(s)$ , the gain-crossover frequency is defined by

$$|L(j\omega_{gc})| = 1 \quad (11)$$

or

$$20 \log_{10} |L(j\omega_{gc})| = 0 \text{ dB}. \quad (12)$$

The phase-crossover frequency is defined by

$$\angle L(j\omega_{pc}) = -180^\circ. \quad (13)$$

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The gain margin is

$$GM = \frac{1}{|L(j\omega_{pc})|} \quad (14)$$

or

$$GM_{dB} = -20 \log_{10} |L(j\omega_{pc})|. \quad (15)$$

The phase margin is

$$PM = 180^\circ + \angle L(j\omega_{gc}). \quad (16)$$

For this experiment, the displayed Bode plots are forward amplifier transfer functions. Hence the unity-gain frequency can be read directly from the plotted gain response, but formal closed-loop gain-margin and phase-margin values require a defined feedback-loop transfer function.

## Section 3: Part A – Resistive-Load CS Amplifier

### 3.1 Circuit and Design Flow

The final Part-A circuit is a common-source NMOS amplifier with resistive drain load. In the Cadence schematic, the driver is  $M_0$ , the drain resistor is  $R_0$ , and the load capacitor is  $C_0$ . Thus,  $R_0 = R_L = 1.52 \text{ k}\Omega$  and  $C_0 = C_L = 5 \text{ pF}$ .

The specified voltage-gain magnitude is

$$|A_V| = 4 \text{ V/V}. \quad (17)$$

A 20% design margin is applied to the gain:

$$A_{V,\text{margin}} = 1.2(4) = 4.8. \quad (18)$$

The transistor width is obtained from the MOS bias / device-characterization circuit and then used in the final CS amplifier.

## Section 4: Part A – Device Characterization and $g_m/I_D$ Sizing

### 4.1 Typed Calculation from the Handwritten Design Sheet

The handwritten calculation is used as the source of the design values. The numerical calculation is reproduced below in typed form.

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$$|A_V| = 4 \quad (19)$$

$$A_{V,\text{margin}} = 1.2(4) = 4.8. \quad (20)$$

The unity-gain frequency is

$$f_u = 100 \text{ MHz} \quad (21)$$

and the load capacitor in the final Cadence schematic is

$$C_0 = C_L = 5 \text{ pF}. \quad (22)$$

Therefore,

$$g_m = 2\pi f_u C_0 \quad (23)$$

$$= 2\pi(100 \times 10^6)(5 \times 10^{-12}) \quad (24)$$

$$= 3.14 \times 10^{-3} \text{ S} \quad (25)$$

$$= 3.14 \text{ mS}. \quad (26)$$

The drain resistor in the final Cadence schematic is

$$R_0 = \frac{A_{V,\text{margin}}}{g_m} \quad (27)$$

$$= \frac{4.8}{3.14 \times 10^{-3}} \quad (28)$$

$$\approx 1.52 \text{ k}\Omega. \quad (29)$$

For

$$V_{DD} = 1.8 \text{ V} \quad (30)$$

and

$$V_O = 0.9 \text{ V}, \quad (31)$$

the drain current is

$$I_D = \frac{V_{DD} - V_O}{R_0} \quad (32)$$

$$= \frac{1.8 - 0.9}{1.52 \text{ k}\Omega} \quad (33)$$

$$\approx 0.592 \text{ mA}. \quad (34)$$

The selected device operating point is

$$\frac{g_m}{I_D} = 5.32888 \text{ V}^{-1}. \quad (35)$$

The characterization plot gives the selected point at

$$V_{GS} \approx 761.2385 \text{ mV}, \quad (36)$$

with

$$\frac{I_D}{W} = 72.32387 \text{ } \mu\text{A}/\mu\text{m}. \quad (37)$$

Hence,

$$W = \frac{I_D}{I_D/W} \quad (38)$$

$$= \frac{0.592 \text{ mA}}{72.32387 \text{ } \mu\text{A}/\mu\text{m}} \quad (39)$$

$$\approx 8.18 \text{ } \mu\text{m}. \quad (40)$$

Therefore,

$$\boxed{W_A \approx 8.18 \text{ } \mu\text{m}} \quad (41)$$

**Table 2:** Part-A design calculation

| Quantity             | Basis                   | Value                              |
|----------------------|-------------------------|------------------------------------|
| Target gain          | $ A_V $                 | 4 V/V                              |
| Gain with 20% margin | $1.2 \times 4$          | 4.8 V/V                            |
| $f_u$                | Specification           | 100 MHz                            |
| $C_0 = C_L$          | Specification           | 5 pF                               |
| $g_m$                | $2\pi f_u C_0$          | 3.14 mS                            |
| $R_0 = R_L$          | $4.8/g_m$               | 1.52 k $\Omega$                    |
| $V_O$                | Bias point              | 0.9 V                              |
| $I_D$                | $(1.8 - 0.9)/R_0$       | 0.592 mA                           |
| $V_{GS}$             | Characterization marker | 761.2385 mV                        |
| $g_m/I_D$            | Characterization marker | 5.32888 V $^{-1}$                  |
| $I_D/W$              | Characterization marker | 72.32387 $\mu\text{A}/\mu\text{m}$ |
| $W$                  | $I_D/(I_D/W)$           | 8.18 $\mu\text{m}$                 |

## 4.2 MOS Bias / Device Characterization

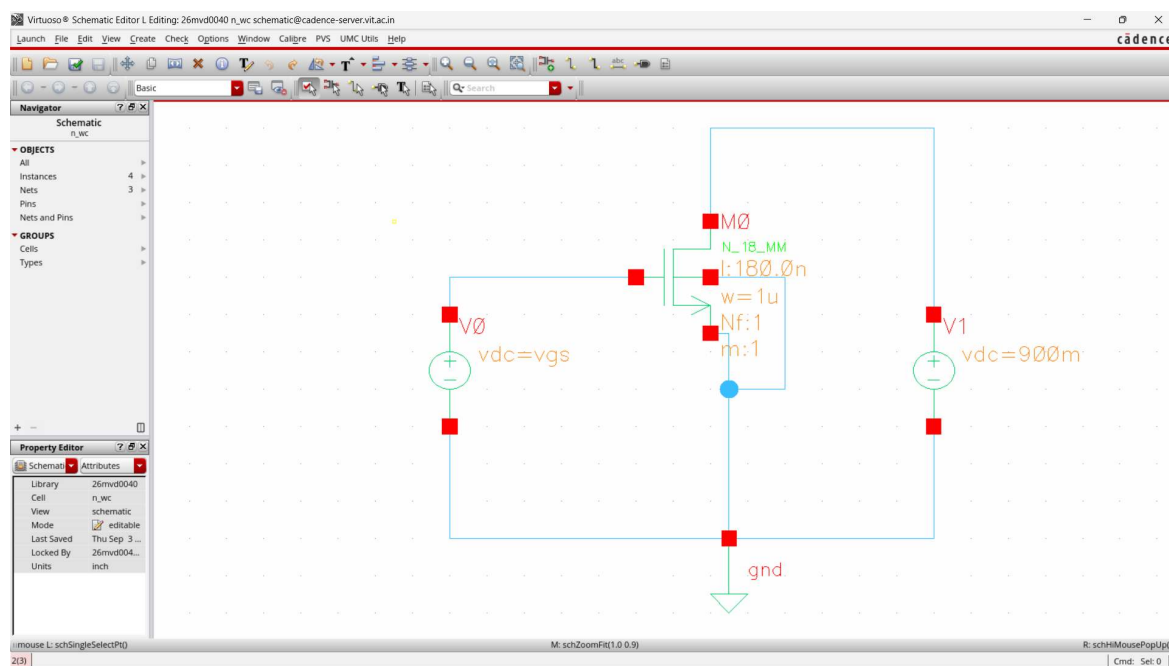


Figure 1: Part A – NMOS bias / device-characterization schematic.

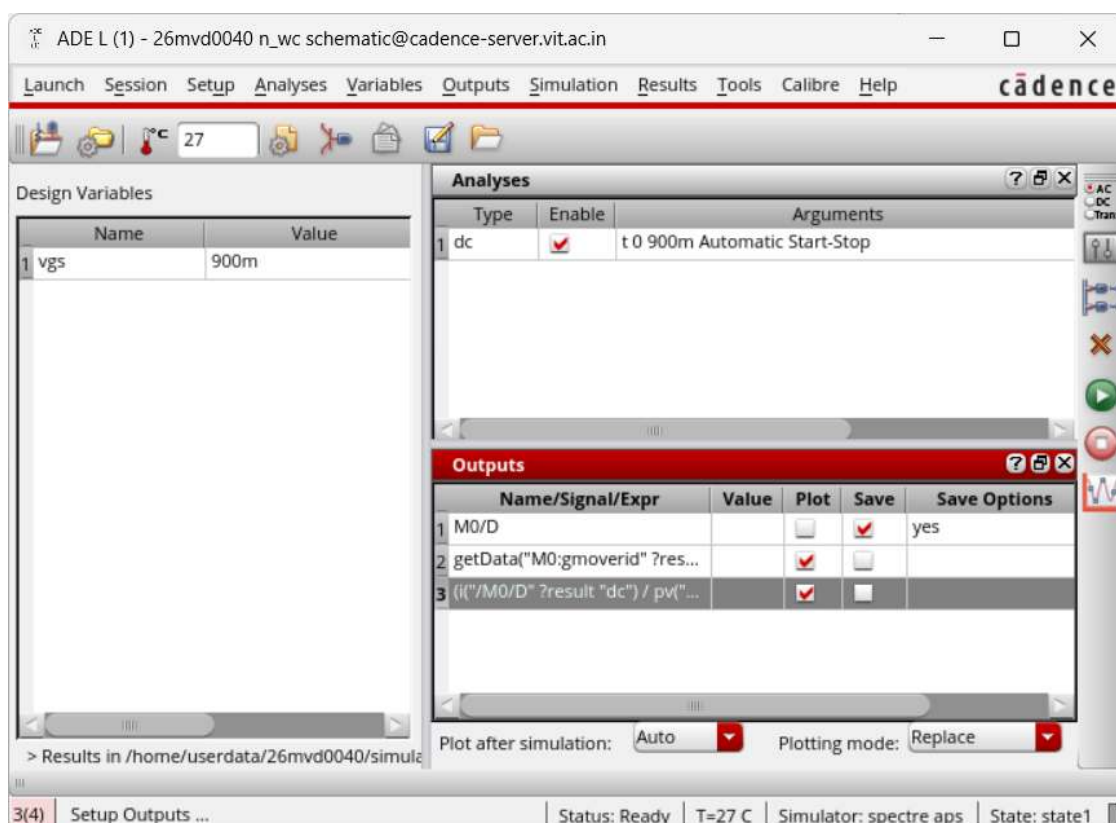
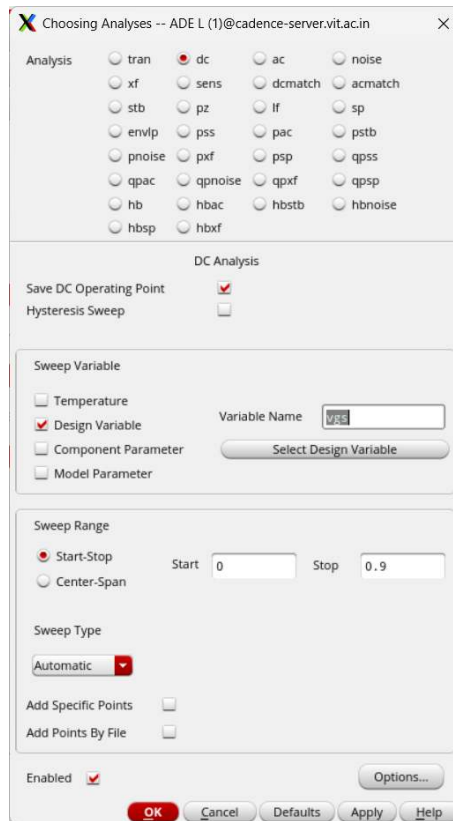
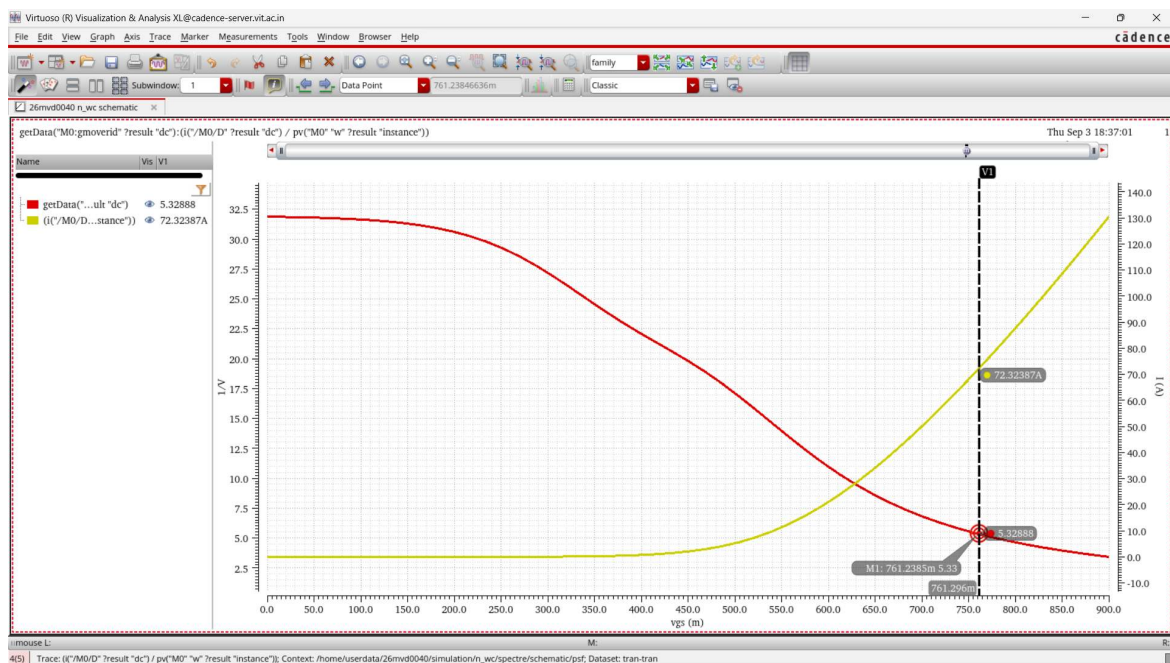


Figure 2: Part A – ADE-L setup for device characterization.

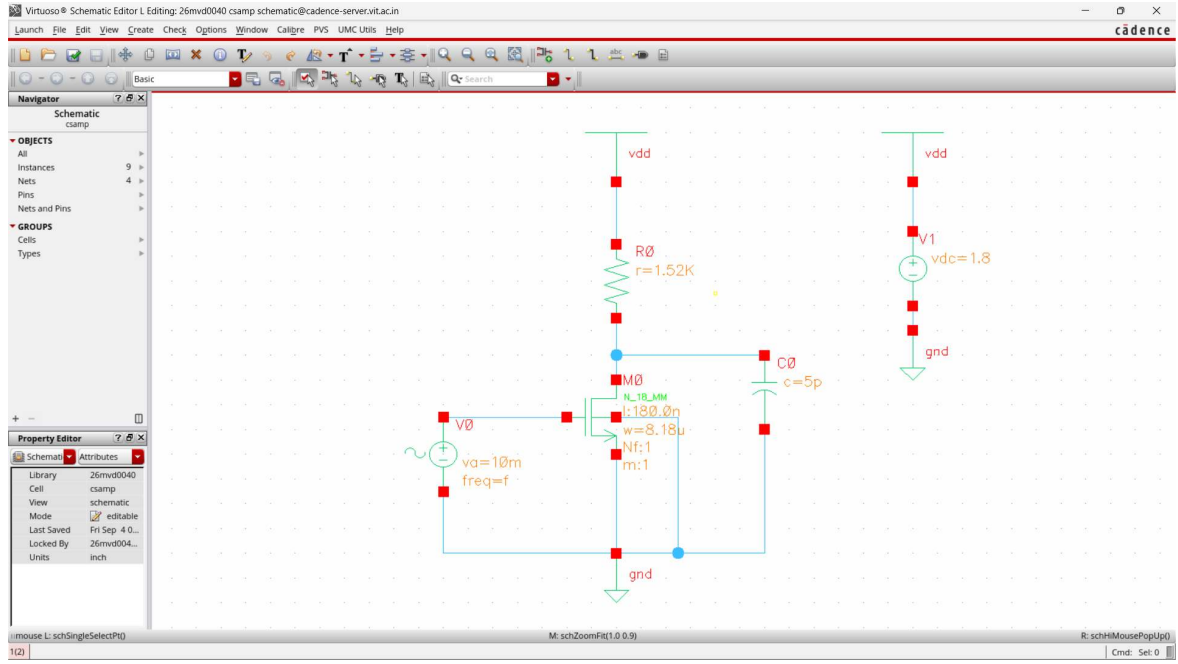


**Figure 3:** Part A – analysis setup used for device characterization.



**Figure 4:** Part A – characterization curves used to obtain the operating point.

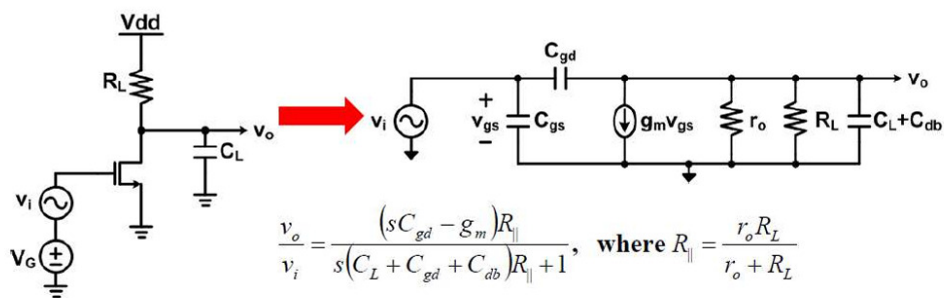
## 4.3 Final Part-A Schematic



**Figure 5:** Part A – final CS amplifier schematic with resistive drain load and sized NMOS.

## Section 5: Part A – Small-Signal Model and Pole-Zero Analysis

### 5.1 Small-Signal Model



**Figure 6:** Small-signal model of the resistive-load common-source amplifier.

The small-signal figure uses  $R_L$  for the load resistor and  $C_L$  for the load capacitance. Its transfer function is

$$\frac{v_o}{v_i} = \frac{(sC_{gd} - g_m)R_\ell}{s(C_L + C_{gd} + C_{db})R_\ell + 1}, \quad (42)$$

where

$$R_\ell = r_o \parallel R_L = \frac{r_o R_L}{r_o + R_L}. \quad (43)$$

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For the actual Cadence schematic,  $R_L$  corresponds to  $R_0 = 1.52 \text{ k}\Omega$  and  $C_L$  corresponds to  $C_0 = 5 \text{ pF}$ ; the driver transistor is  $M_0$ .

From the same transfer function, the first-order pole and zero are

$$\omega_{pA} = \frac{1}{R_\ell(C_L + C_{gd} + C_{db})}, \quad \omega_{zA} = \frac{g_m}{C_{gd}}. \quad (44)$$

The simulated pole-zero values below are the actual extracted values for the implemented circuit.

## 5.2 Measured Pole-Zero Data

The supplied pole-zero simulation gives

$$f_{pA} \approx -2.53824 \times 10^7 \text{ Hz} \quad (45)$$

and

$$f_{zA} \approx 1.53047 \times 10^{11} \text{ Hz}. \quad (46)$$

The DC gain reported by the pole-zero analysis is

$$\boxed{|A_{DC,A}| = 3.84258 \text{ V/V}}. \quad (47)$$

Its corresponding gain in decibels is

$$A_{DC,A,\text{dB}} = 20 \log_{10}(3.84258) \quad (48)$$

$$\approx 11.69 \text{ dB}. \quad (49)$$

Using the extracted pole and zero, a compact measured representation is

$$H_A(s) \approx -3.84258 \frac{1 - s/\omega_{zA}}{1 + s/\omega_{pA}}, \quad (50)$$

where

$$\omega_{pA} = 2\pi(25.3824 \text{ MHz}) \quad (51)$$

and

$$\omega_{zA} = 2\pi(153.047 \text{ GHz}). \quad (52)$$

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## Section 6: Part A – Simulated Gain, $-3$ dB Response and Unity-Gain Frequency

### 6.1 Design Gain versus Simulated DC Gain

The design target was

$$\boxed{|A_V|_{\text{target}} = 4 \text{ V/V}}. \quad (53)$$

After simulation, the pole-zero analysis gives

$$\boxed{|A_V|_{\text{sim}} = 3.84258 \text{ V/V}}. \quad (54)$$

The target gain in decibels is

$$A_{V,\text{target,dB}} = 20 \log_{10}(4) \quad (55)$$

$$\approx 12.04 \text{ dB}. \quad (56)$$

The simulated DC gain is

$$A_{V,\text{sim,dB}} = 20 \log_{10}(3.84258) \quad (57)$$

$$\approx 11.69 \text{ dB}. \quad (58)$$

Therefore,

$$\Delta A_V = 12.04 - 11.69 \quad (59)$$

$$\approx 0.35 \text{ dB}. \quad (60)$$

The percentage difference in voltage gain is

$$\% \text{difference} = \frac{4 - 3.84258}{4} \times 100 \quad (61)$$

$$\approx 3.94\%. \quad (62)$$

Hence, the simulation produces approximately  $3.84 \text{ V/V}$  compared with the required  $4 \text{ V/V}$ , corresponding to a deviation of about  $3.94\%$ .

**Table 3:** Part-A gain comparison

| Parameter             | Value       |
|-----------------------|-------------|
| Specified gain        | 4.000 V/V   |
| Specified gain        | 12.04 dB    |
| Simulated DC gain     | 3.84258 V/V |
| Simulated DC gain     | 11.69 dB    |
| Difference            | 0.35 dB     |
| Percentage difference | 3.94%       |

## 6.2 Measured $-3$ dB and Unity-Gain Response

The supplied Cadence frequency-response plots are used directly.

The unity-gain marker gives

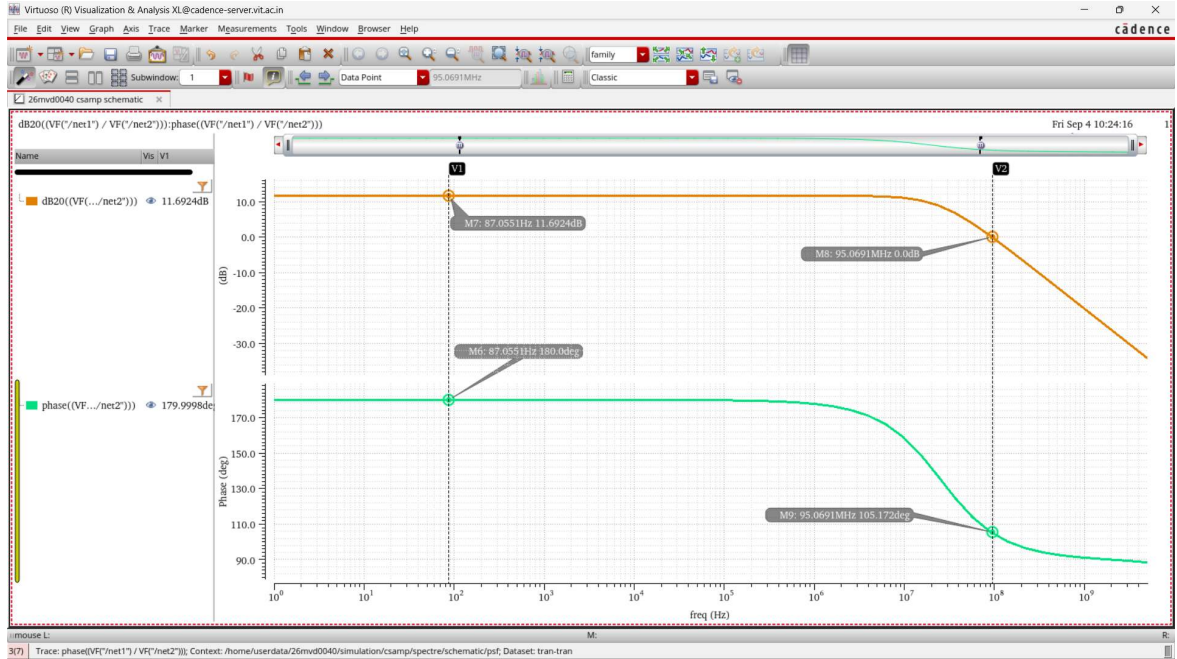
$$f_{gc,A} = 95.0691 \text{ MHz}, \quad (63)$$

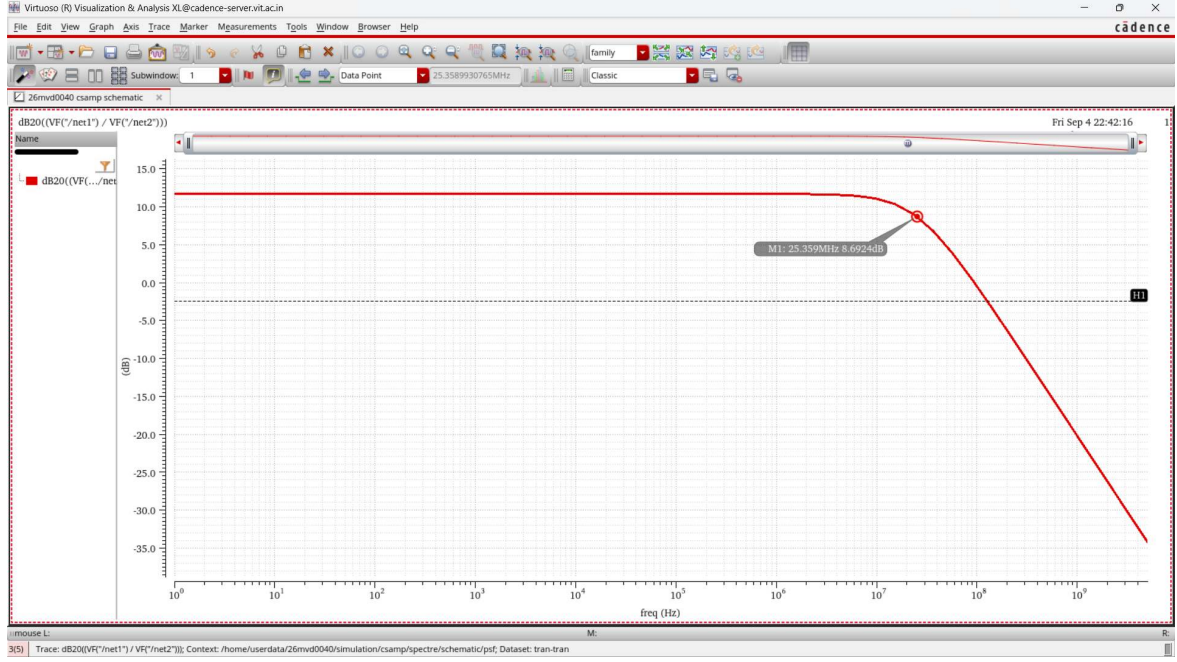
at 0 dB. The phase marker at this same frequency is approximately  $105.172^\circ$  in the plotted phase convention.

The supplied high-frequency response gives the  $-3$  dB point at

$$f_{3dB,A} = 25.3599 \text{ MHz}, \quad (64)$$

where the marked gain is approximately 8.6924 dB.

**Figure 7:** Part A – gain and phase response with the unity-gain marker.



**Figure 8:** Part A – final gain response showing high-frequency roll-off and the  $-3$  dB marker.

## Section 7: Part A – Gain Margin and Phase Margin

For a feedback-loop transfer function  $L(s)$ ,

$$|L(j\omega_{gc})| = 1 \quad (65)$$

defines the gain-crossover frequency, while

$$\angle L(j\omega_{pc}) = -180^\circ \quad (66)$$

defines the phase-crossover frequency.

The gain margin is

$$GM_{dB} = -20 \log_{10} |L(j\omega_{pc})| \quad (67)$$

and the phase margin is

$$PM = 180^\circ + \angle L(j\omega_{gc}). \quad (68)$$

From the supplied Part-A gain/phase plot, the 0 dB crossing is

$$f_{gc,A} = 95.0691 \text{ MHz}. \quad (69)$$

The corresponding phase marker is approximately  $105.172^\circ$  at 95.0691 MHz in the plotted phase convention. This is the gain-crossover frequency of the displayed forward amplifier magnitude.

The  $-3$  dB frequency,

$$f_{3\text{dB},A} = 25.3599 \text{ MHz}, \quad (70)$$

is the amplifier bandwidth marker and should not be confused with the gain-crossover frequency.

Because the supplied plot is the forward transfer function rather than a complete feedback-loop gain plot, a formal numerical gain margin and phase margin are not claimed. A numerical phase margin requires the phase of the loop transfer function at the gain-crossover frequency, while a numerical gain margin requires a phase-crossover frequency at which the loop phase reaches  $-180^\circ$ .

## 7.1 Part-A Simulation Record

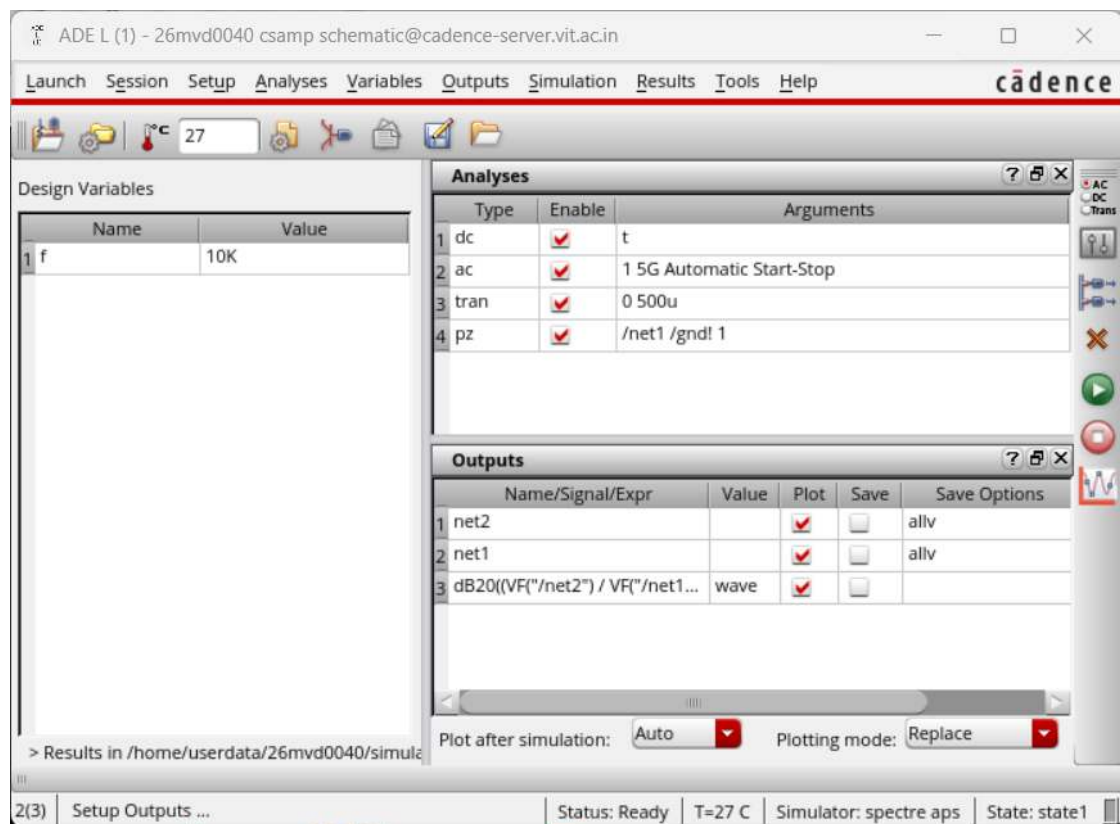


Figure 9: Part A – ADE-L simulation setup.

```

DC simulation time: CPU = 68 us, elapsed = 67.9493 us.
DC simulation time: CPU = 30 us, elapsed = 30.0407 us.
      Poles (Hz)

      Real              Imaginary          Qfactor
1  -2.53824e+07         0.00000e+00         5.00000e-01

      Zeros (Hz)
      at V(net1,0)/V0

      Real              Imaginary          Qfactor
1   1.53047e+11         0.00000e+00        -5.00000e-01

Constant factor = 6.37282e-04
DC gain = 3.84258e+00

```

Figure 10: Part A – pole-zero analysis output.

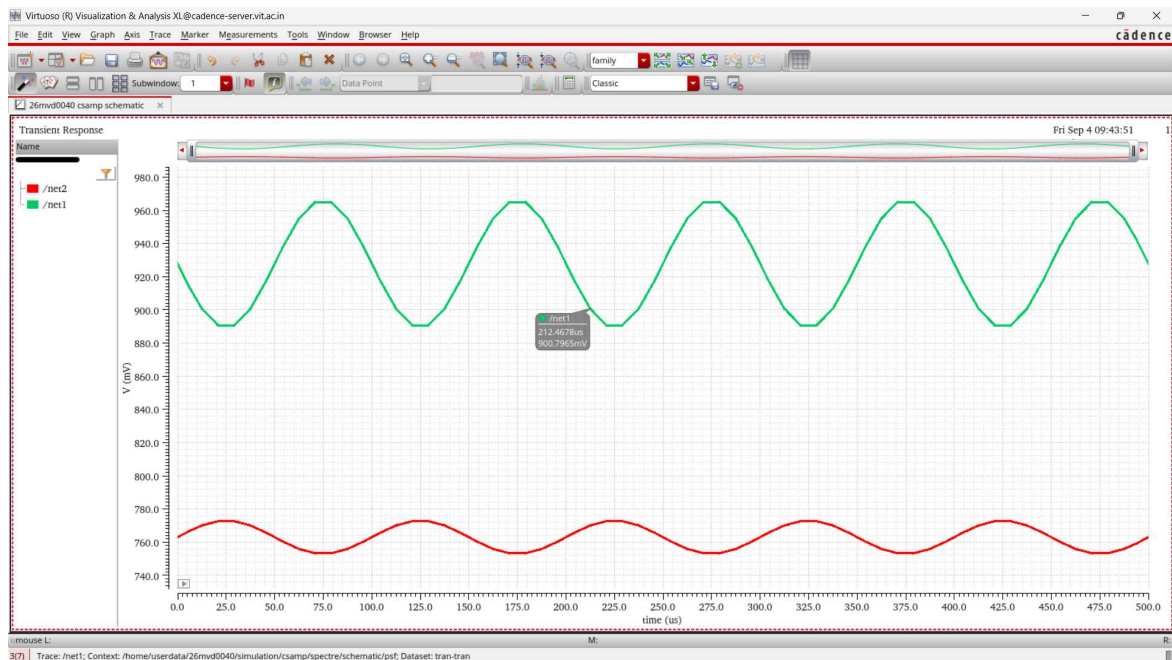


Figure 11: Part A – transient response.

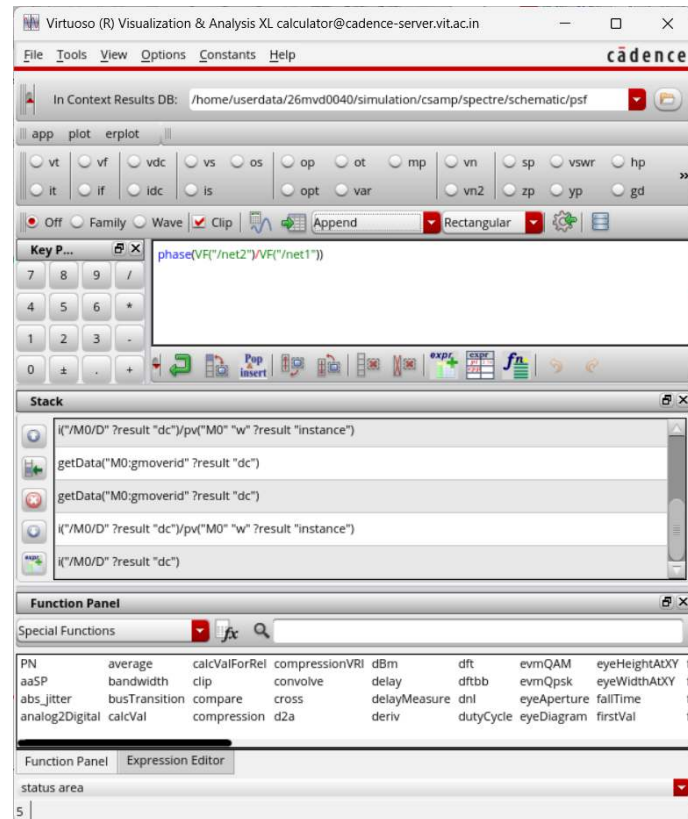


Figure 12: Part A – calculator / expression window.

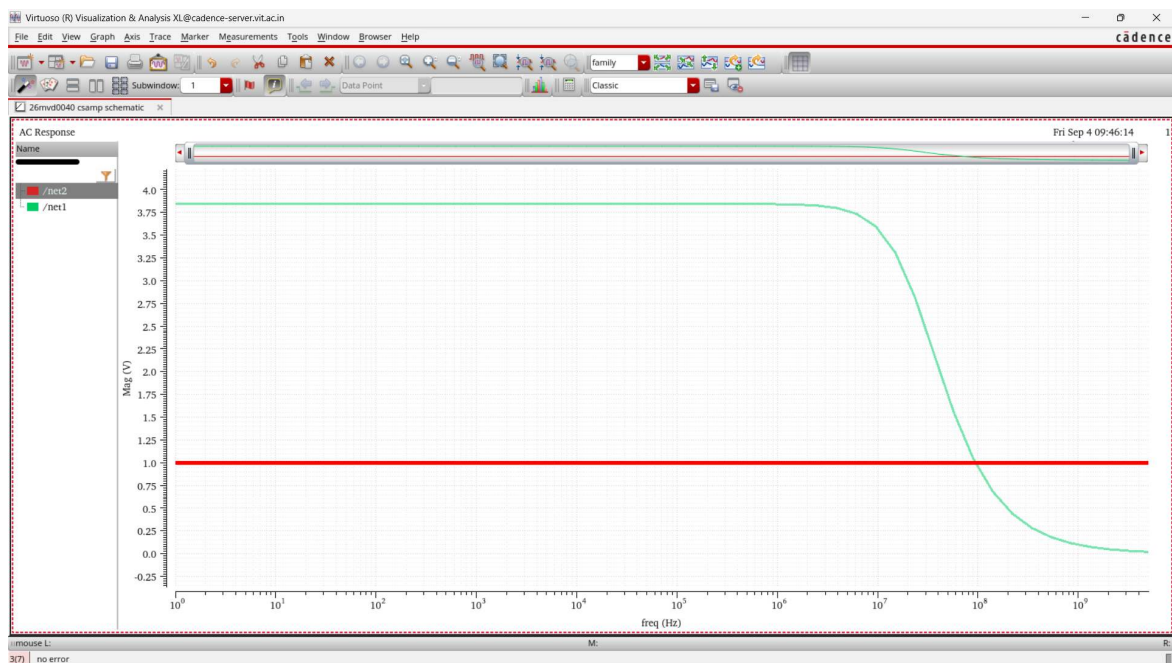


Figure 13: Part A – DC transfer characteristic.

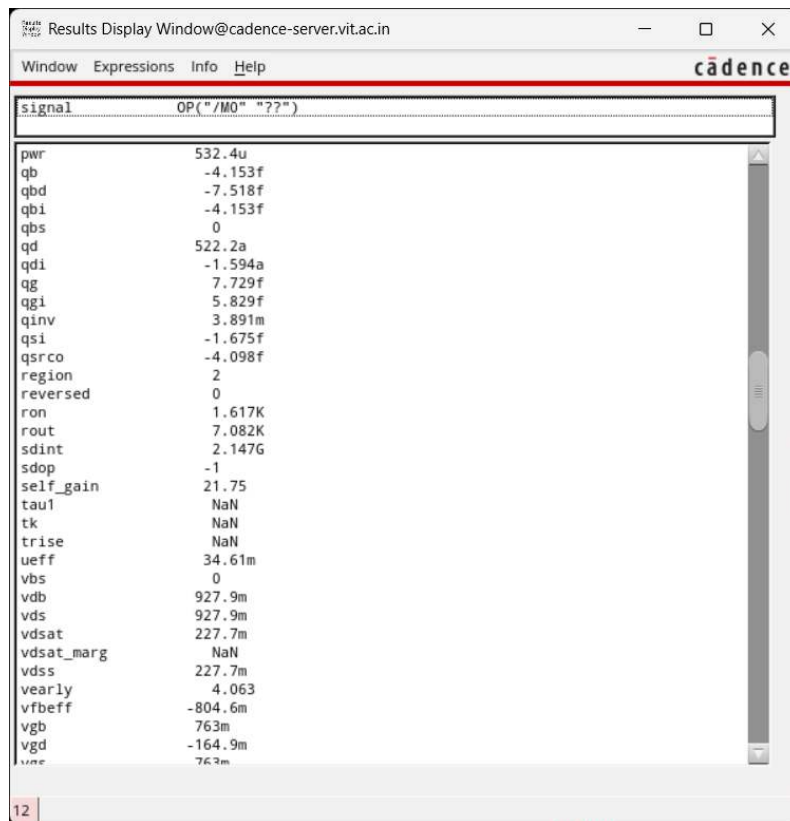


Figure 14: Part A – operating-point result.

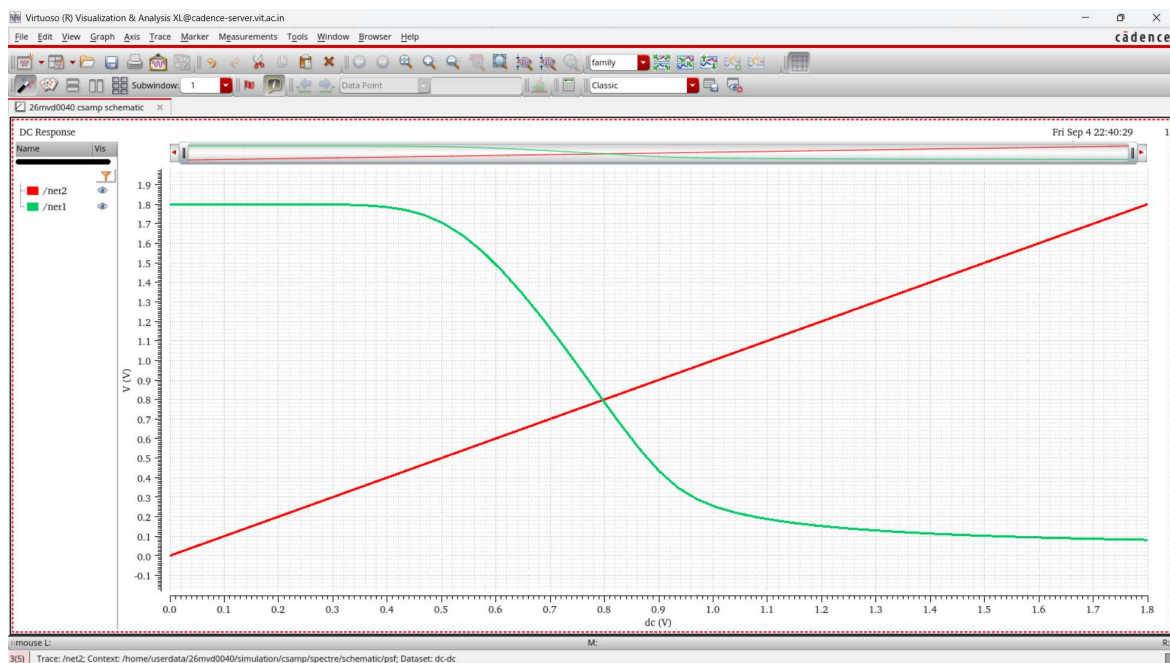


Figure 15: Part A – AC-response plot used for verification.

## Section 8: Part B – Current-Mirror / Active-Load CS Amplifier

### 8.1 Main Active-Load CS Amplifier

Part B uses an NMOS common-source driver with a PMOS active-load network. In the final Cadence schematic, the NMOS driver is  $M_0$  and the PMOS active-load device is  $M_1$ . The uploaded textbook small-signal figure uses NMOS  $M_1$  and PMOS  $M_2$ ; the variable mapping is stated explicitly below.

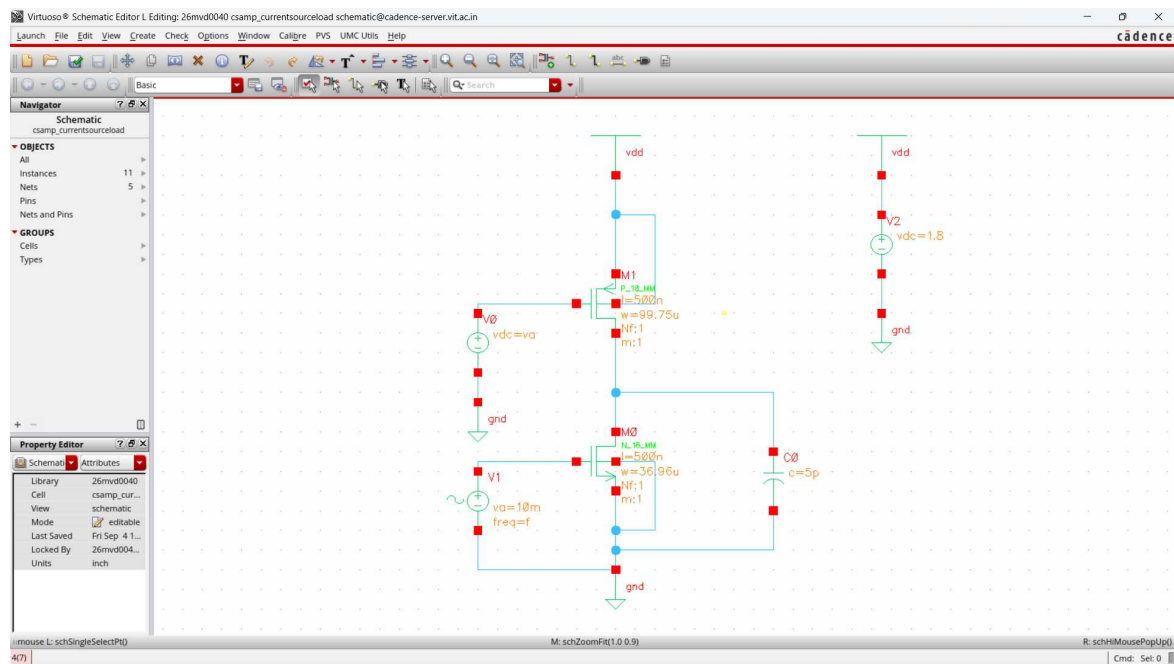
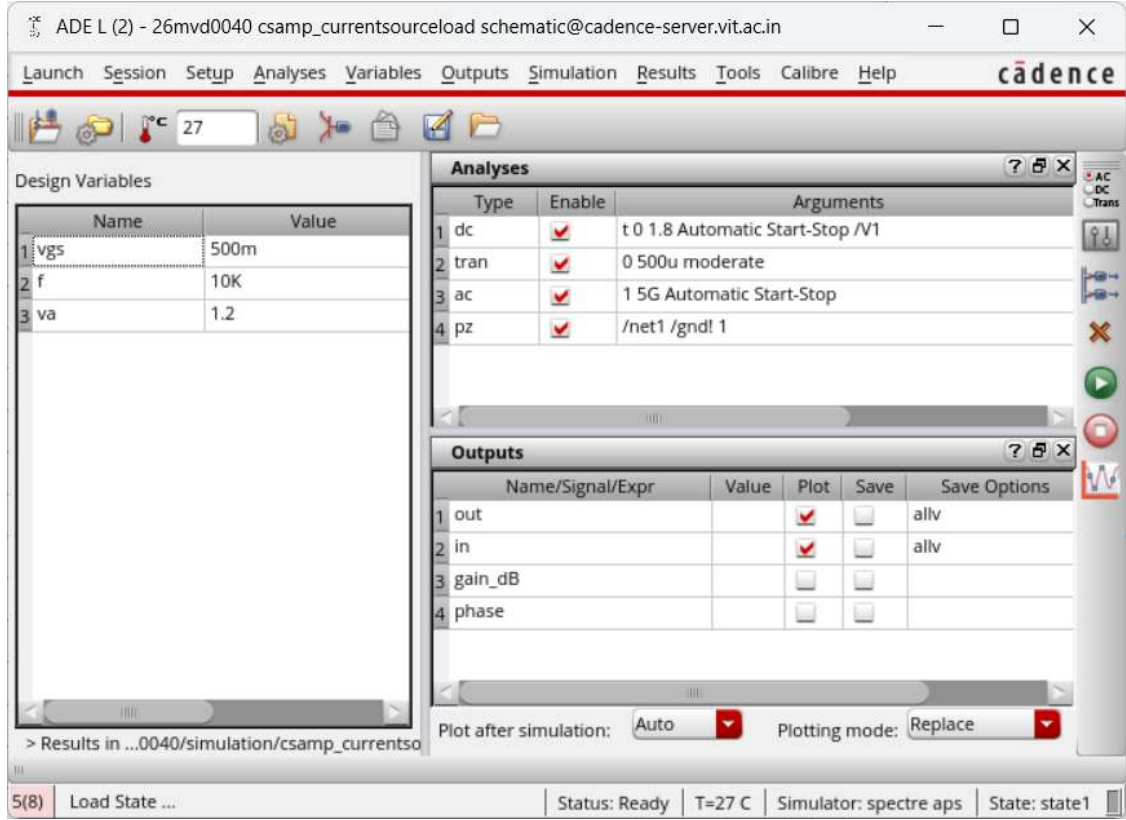


Figure 16: Part B – complete current-mirror / active-load CS amplifier schematic.



**Figure 17:** Part B – ADE-L setup associated with the active-load CS amplifier.

## 8.2 Design Equations and Power Constraint

The design requirements are

$$|A_V| = 4 \text{ V/V}, \quad (71)$$

$$f_u = 100 \text{ MHz}, \quad (72)$$

$$C_L = 5 \text{ pF}, \quad (73)$$

and

$$P_{DC} = 1 \text{ mW}. \quad (74)$$

The power relation is

$$I_{DC} = \frac{P_{DC}}{V_{DD}}. \quad (75)$$

For  $V_{DD} = 1.8 \text{ V}$ ,

$$I_{DC} = \frac{1 \text{ mW}}{1.8 \text{ V}} \quad (76)$$

$$\approx 0.556 \text{ mA}. \quad (77)$$

The supplied current-source design uses approximately

$$I_D = 0.6 \text{ mA}. \quad (78)$$

At  $V_{DD} = 1.8$  V, this corresponds to approximately 1.08 mW. The 0.6 mA value is therefore the current used in the supplied current-source calculation and is slightly above the nominal 1 mW target.

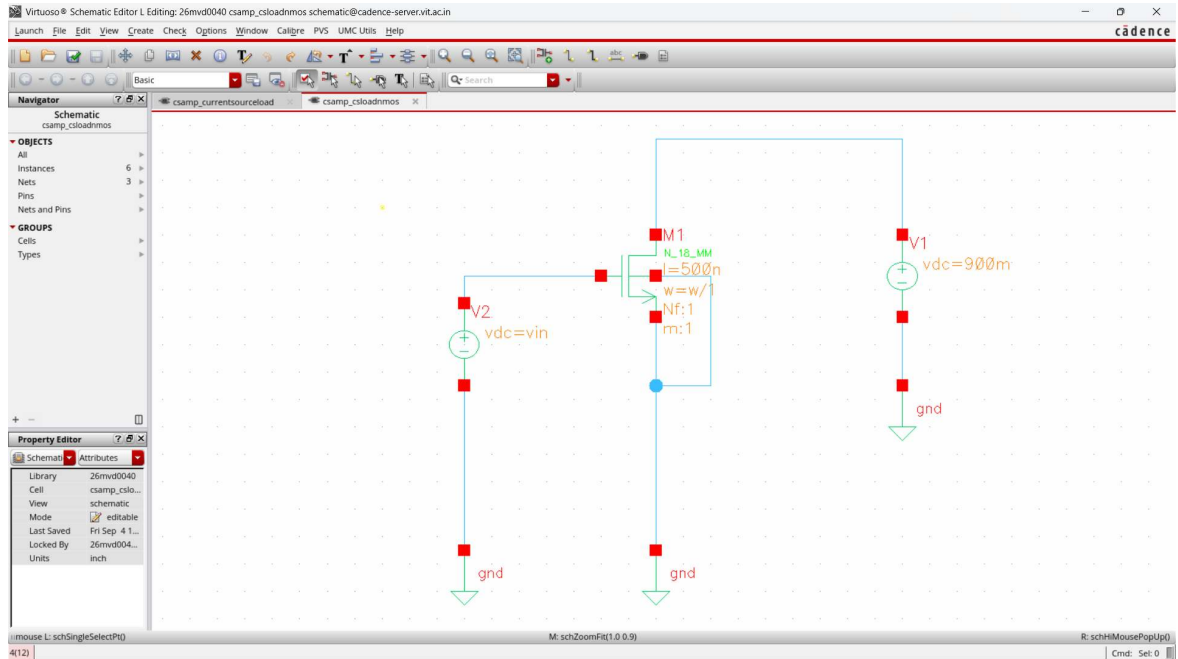
The required transconductance is

$$g_m = 2\pi f_u C_L. \quad (79)$$

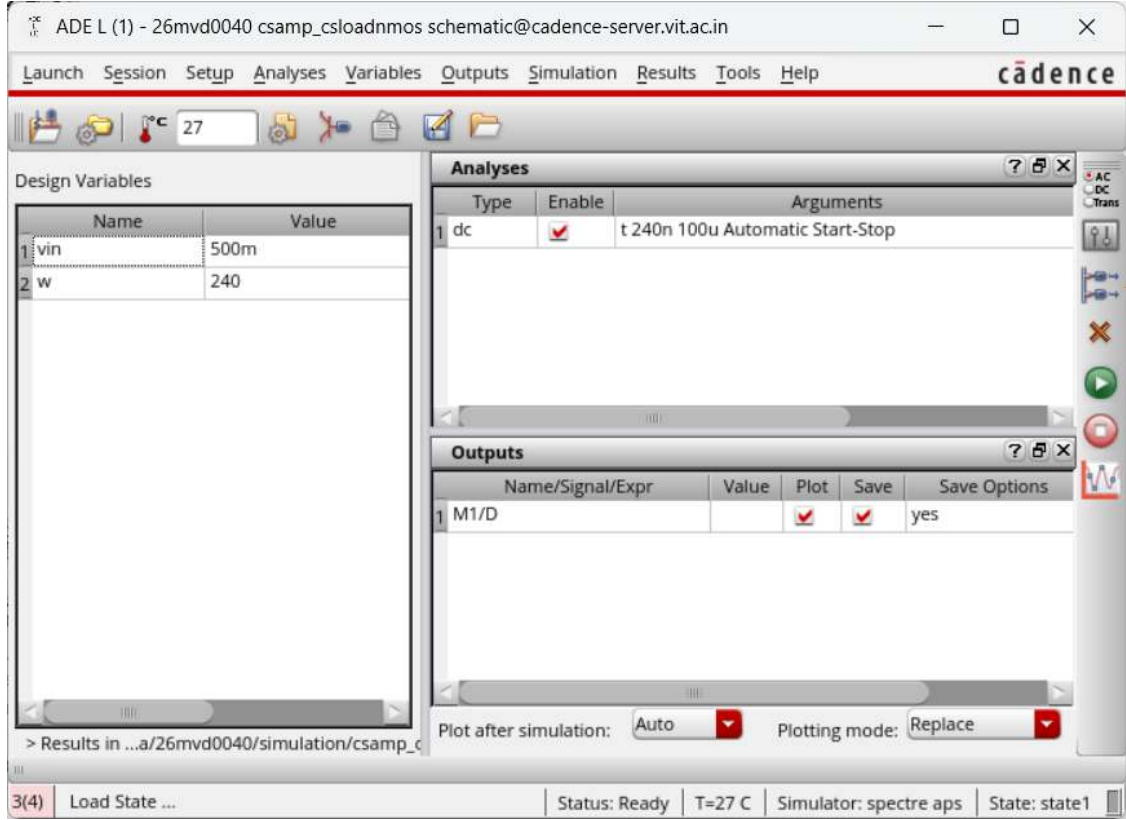
## Section 9: Part B – Separate NMOS and PMOS Sizing by W Sweep

### 9.1 NMOS Driver Width Calculation

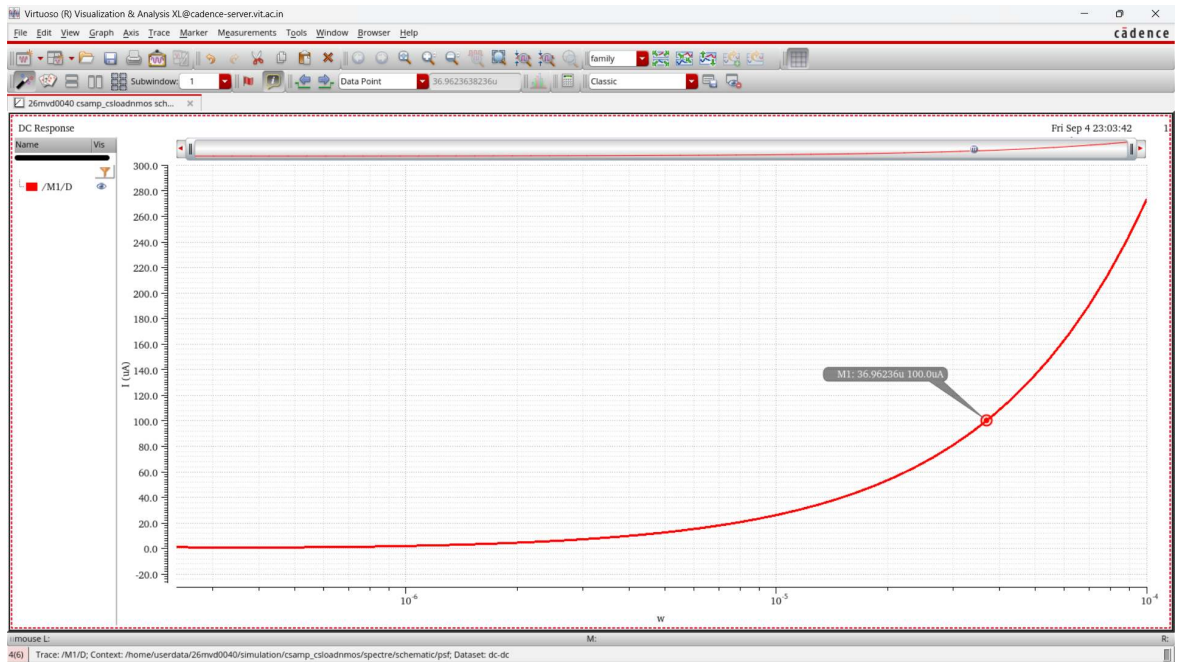
The NMOS driver is characterized independently.



**Figure 18:** Part B – NMOS device-characterization schematic.



**Figure 19:** Part B – NMOS design-variable setup for the W sweep.



**Figure 20:** Part B – NMOS current-versus-width sweep used to determine the driver width.

From the supplied sweep marker, the selected point is  $I_D \approx +100 \mu\text{A}$ . Therefore,

$$W_N \approx 36.96236 \mu\text{m}. \quad (80)$$

This width is taken directly from the actual  $100\text{-}\mu\text{A}$  sweep.

## 9.2 PMOS Active-Load Width Calculation

The PMOS transistor is characterized independently because its current-density characteristic differs from the NMOS device.

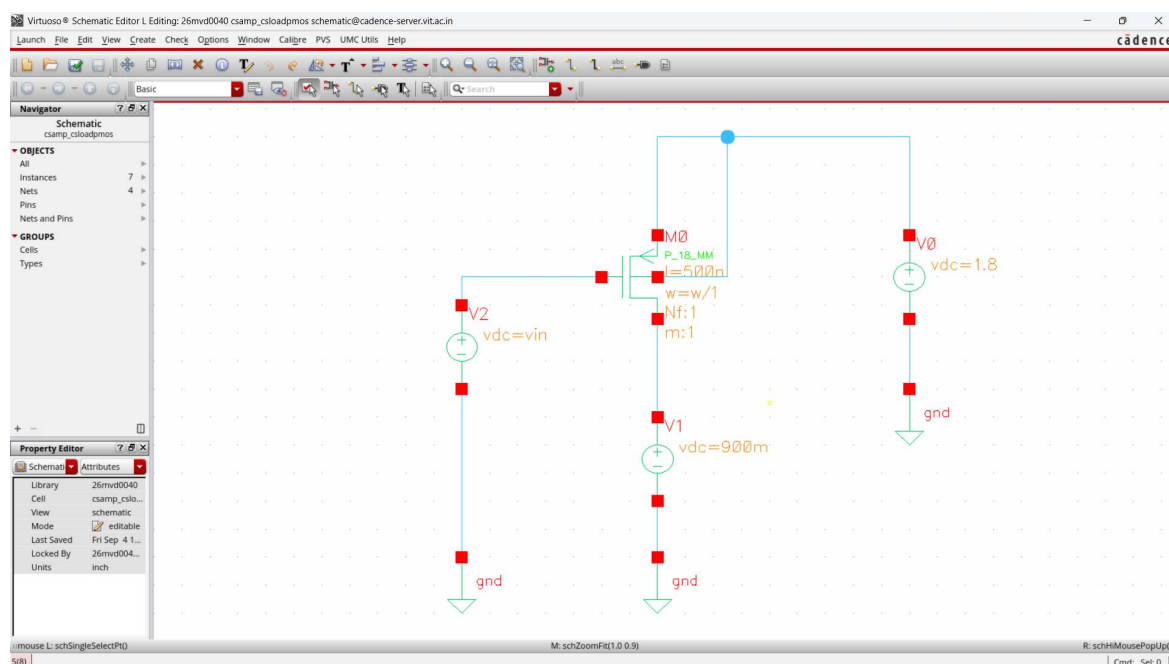


Figure 21: Part B – PMOS device-characterization / bias schematic.

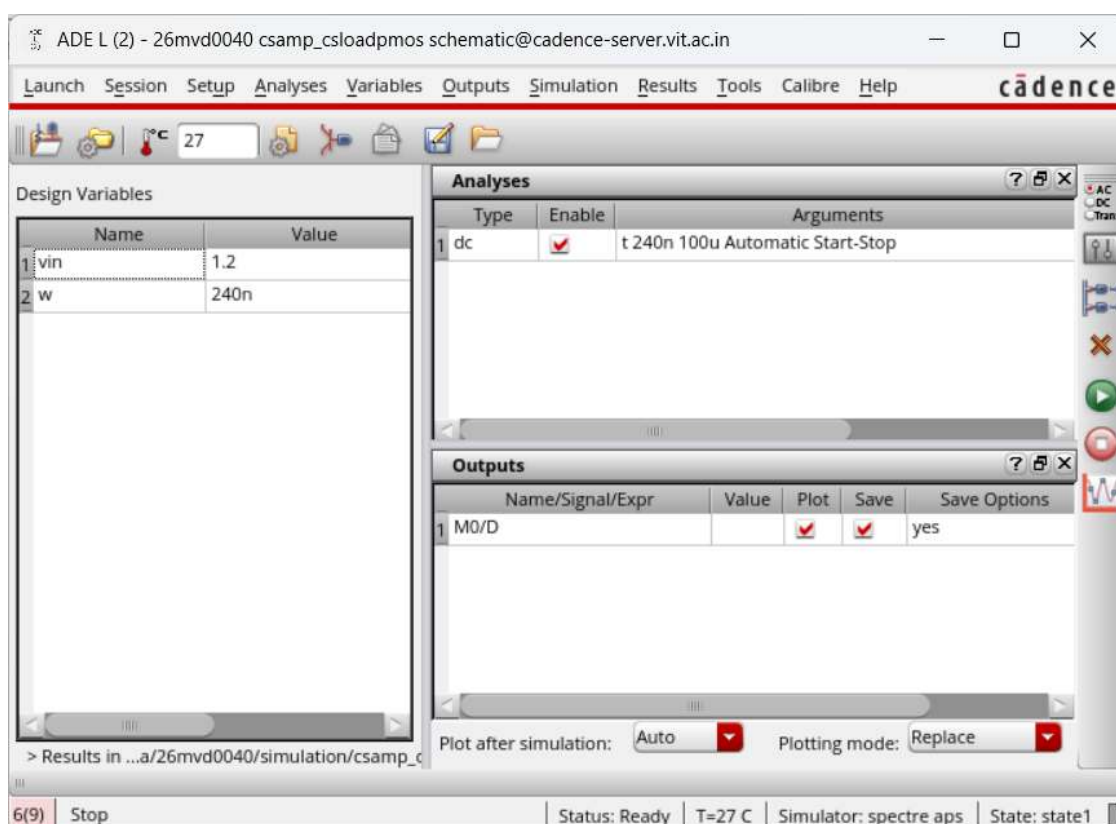
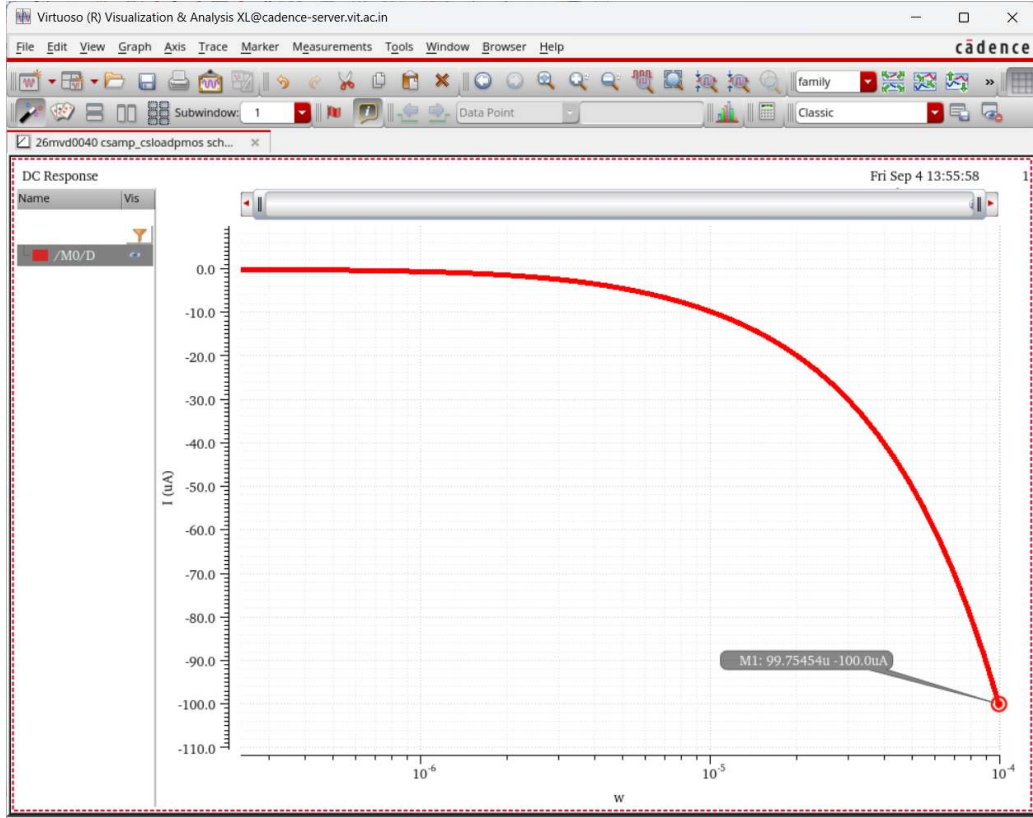


Figure 22: Part B – PMOS design-variable setup for the W sweep.



**Figure 23:** Part B – PMOS current-versus-width sweep used to determine the active-load width.

From the supplied sweep marker, the selected point is  $I_D \approx -100 \mu\text{A}$ . Therefore,

$$W_P \approx 99.75454 \mu\text{m}. \quad (81)$$

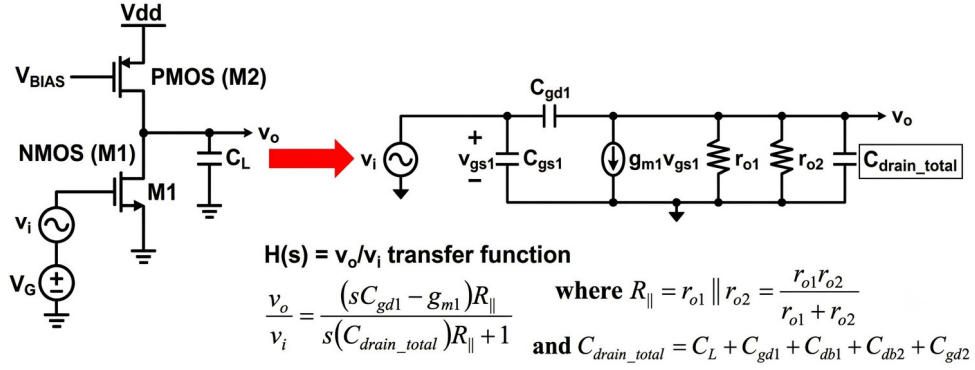
This width is taken directly from the actual  $-100 \mu\text{A}$  sweep.

**Table 4:** Part-B transistor widths

| Device           | Sweep condition                | Selected width         |
|------------------|--------------------------------|------------------------|
| NMOS driver      | $I_D \approx 100 \mu\text{A}$  | $36.96236 \mu\text{m}$ |
| PMOS active load | $I_D \approx -100 \mu\text{A}$ | $99.75454 \mu\text{m}$ |

## Section 10: Part B – Small-Signal Model and Transfer Function

### 10.1 Small-Signal Model



**Figure 24:** Small-signal model of the CS amplifier with PMOS active load.

The small-signal figure labels the NMOS driver as  $M_1$  and the PMOS active-load device as  $M_2$ . For that model,

$$R_{\parallel} = r_{o1} \parallel r_{o2} = \frac{r_{o1}r_{o2}}{r_{o1} + r_{o2}}, \quad (82)$$

and

$$C_{drain,total} = C_L + C_{gd1} + C_{db1} + C_{db2} + C_{gd2}. \quad (83)$$

The transfer function shown in the figure is

$$\boxed{\frac{v_o}{v_i} = \frac{(sC_{gd1} - g_{m1})R_{\parallel}}{sC_{drain,total}R_{\parallel} + 1}}. \quad (84)$$

For the actual Cadence final schematic, the NMOS driver is  $M_0$  and the PMOS active-load device is  $M_1$ . Therefore,

$$M_1 \text{ (textbook)} \equiv M_0 \text{ (Cadence)}, \quad M_2 \text{ (textbook)} \equiv M_1 \text{ (Cadence)}. \quad (85)$$

Using the Cadence device labels,

$$\boxed{H_B(s) = \frac{(sC_{gd0} - g_{m0})(r_{o0} \parallel r_{o1})}{sC_{drain,total}(r_{o0} \parallel r_{o1}) + 1}}, \quad (86)$$

where

$$C_{drain,total} = C_0 + C_{gd0} + C_{db0} + C_{db1} + C_{gd1}. \quad (87)$$

At frequencies well below the zero, the gain is approximately

$$A_{v0} \approx -g_{m0}(r_{o0} \parallel r_{o1}). \quad (88)$$

For the corresponding first-order form, the pole and zero are

$$\omega_{pB} = \frac{1}{(r_{o0} \parallel r_{o1})C_{\text{drain,total}}}, \quad \omega_{zB} = \frac{g_{m0}}{C_{gd0}}. \quad (89)$$

Additional high-frequency behaviour is determined by the parasitic capacitances represented in  $C_{\text{drain,total}}$ .

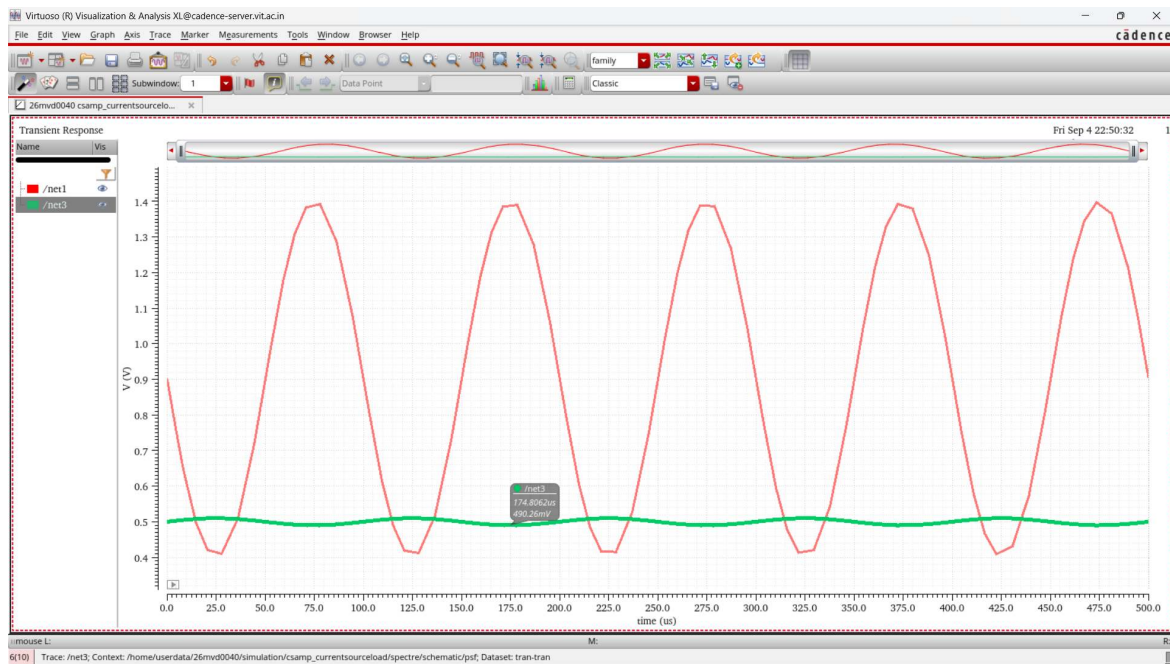
## Section 11: Part B – DC, Transient, AC and Pole-Zero Analysis

### 11.1 DC Analysis



**Figure 25:** Part B – DC transfer response of the active-load CS amplifier.

## 11.2 Transient Analysis



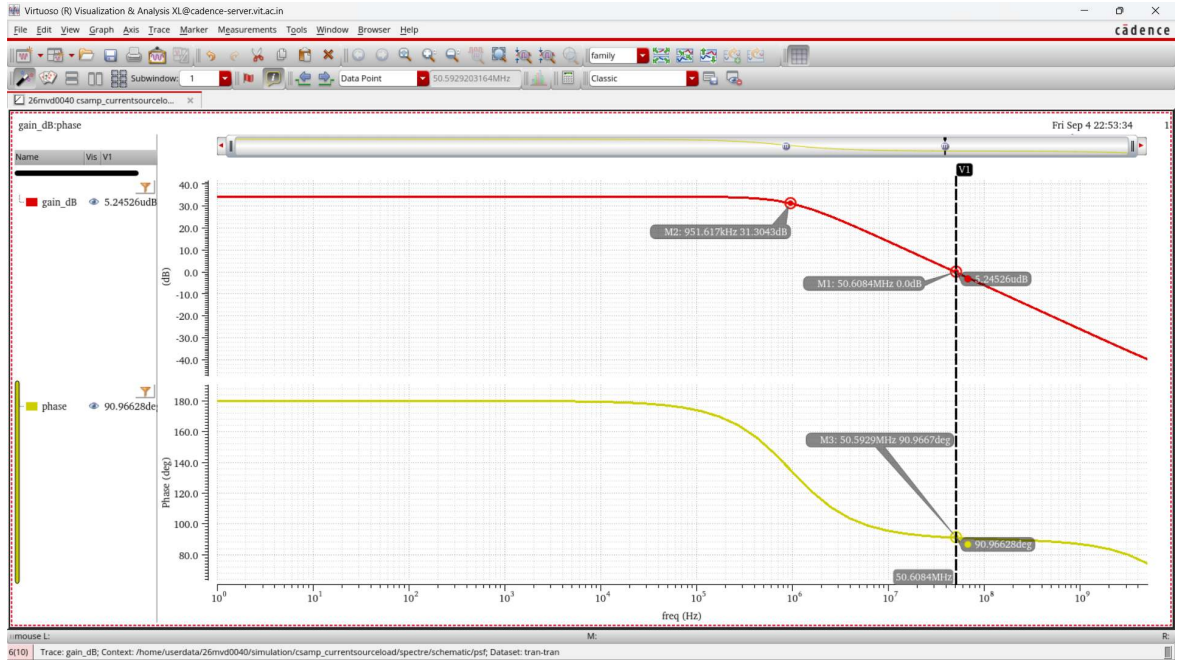
**Figure 26:** Part B – transient response of the active-load CS amplifier.

## 11.3 AC Analysis



**Figure 27:** Part B – AC magnitude response of the active-load CS amplifier.

## 11.4 Gain and Phase Response



**Figure 28:** Part B – AC gain and phase response with unity-gain marker.

## 11.5 Pole-Zero Analysis

```
*****
PZ Analysis 'pz'
*****

Warning from spectre during PZ analysis 'pz'.
WARNING: BSIM3v3 MOS Transistor - frequency dependent components are present in the circuit, approximated as AC equivalents at 1.000000e+00Hz for pz analysis.

DC simulation time: CPU = 66 us, elapsed = 67.9493 us.
DC simulation time: CPU = 38 us, elapsed = 38.147 us.
Poles (Hz)

      Real          Imaginary      Qfactor
1  -9.55158e+05      0.00000e+00      5.00000e-01

Zeros (Hz)
at V(net1,0)/V1

      Real          Imaginary      Qfactor
1  1.77714e+10      0.00000e+00     -5.00000e-01

Constant factor = 2.78980e-03
DC gain = 5.19063e+01

Accumulated DC solution time = 4.19899 s.
Intrinsic pz analysis time = -6.28831 s.
Total time required for pz analysis 'pz': CPU = 2.291 ms, elapsed = 2.30217 ms, util. = 99.5%.
Time accumulated: CPU = 174.373 ms, elapsed = 2.2018 s.
Peak resident memory used = 165 Mbytes.
```

**Figure 29:** Part B – pole-zero analysis output showing the extracted pole, zero and DC gain.

The supplied pole-zero analysis includes a Cadence warning that frequency-dependent components are approximated as AC equivalents at 0 Hz for the PZ calculation. With that qualification, the reported pole and zero are

$$f_{pB} = -9.55158 \times 10^5 \text{ Hz} \quad (90)$$

and

$$f_{zB} = 1.77714 \times 10^{10} \text{ Hz}. \quad (91)$$

---

The pole-zero analysis reports

$$\boxed{|A_{DC,B}| = 51.9063 \text{ V/V}}. \quad (92)$$

The corresponding gain in decibels is

$$A_{DC,B,\text{dB}} = 20 \log_{10}(51.9063) \quad (93)$$

$$\approx 34.30 \text{ dB}. \quad (94)$$

Thus, the extracted transfer function is

$$H_B(s) \approx -51.9063 \frac{1 - s/\omega_{zB}}{1 + s/\omega_{pB}}, \quad (95)$$

with

$$\omega_{pB} = 2\pi(0.955158 \text{ MHz}) \quad (96)$$

and

$$\omega_{zB} = 2\pi(17.7714 \text{ GHz}). \quad (97)$$

## Section 12: Part B – Simulated Gain, $-3$ dB Response and Unity-Gain Frequency

### 12.1 Simulated Gain

The Part-B active-load stage was designed for a gain of 4 V/V.

The final pole-zero simulation reports

$$\boxed{|A_{DC,B}| = 51.9063 \text{ V/V}}. \quad (98)$$

The gain in decibels is

$$\boxed{A_{DC,B} \approx 34.30 \text{ dB}}. \quad (99)$$

Therefore, the active-load CS amplifier produces a gain significantly greater than 20 dB.

The gain target of 4 V/V corresponds to

$$20 \log_{10}(4) \approx 12.04 \text{ dB}, \quad (100)$$

whereas the simulated active-load gain is approximately 34.30 dB.

Hence,

$$34.30 \text{ dB} > 20 \text{ dB} \quad (101)$$

and the required gain level is comfortably exceeded.

---

**Table 5:** Part-B simulated gain

| Quantity               | Value              |
|------------------------|--------------------|
| Specified gain         | 4 V/V              |
| Specified gain in dB   | 12.04 dB           |
| Simulated DC gain      | 51.9063 V/V        |
| Simulated DC gain      | 34.30 dB           |
| Gain relative to 20 dB | Greater than 20 dB |

## 12.2 Marked $-3$ dB and Unity-Gain Frequencies

The actual Cadence gain and phase plot shown in Fig. 28 is used directly for the frequency-response analysis.

The marked  $-3$  dB point is

$$f_{3\text{dB},B} \approx 951.617 \text{ kHz} \quad (102)$$

with a corresponding gain of approximately

$$31.3043 \text{ dB}. \quad (103)$$

The unity-gain marker occurs at

$$f_{u,B} \approx 50.6084 \text{ MHz} \quad (104)$$

where

$$A_v = 0 \text{ dB}. \quad (105)$$

**Table 6:** Part-B marked frequency-response values

| Parameter                   | Measured value                |
|-----------------------------|-------------------------------|
| Low-frequency gain          | $\approx 34.30 \text{ dB}$    |
| $-3$ dB marker frequency    | $\approx 951.617 \text{ kHz}$ |
| Gain at $-3$ dB marker      | $\approx 31.3043 \text{ dB}$  |
| Unity-gain frequency        | $\approx 50.6084 \text{ MHz}$ |
| Gain at unity-gain crossing | $0 \text{ dB}$                |

The  $-3$  dB frequency represents the amplifier bandwidth, while the unity-gain frequency represents the frequency at which the amplifier magnitude reaches 0 dB.

---

## Section 13: Part B – Gain Margin and Phase Margin

For a feedback-loop transfer function  $L(s)$ , the gain-crossover frequency is defined by

$$|L(j\omega_{gc})| = 1 \quad (106)$$

and the phase-crossover frequency is defined by

$$\angle L(j\omega_{pc}) = -180^\circ. \quad (107)$$

The gain margin is

$$GM_{dB} = -20 \log_{10} |L(j\omega_{pc})| \quad (108)$$

and the phase margin is

$$PM = 180^\circ + \angle L(j\omega_{gc}). \quad (109)$$

For the Part-B gain and phase plot in Fig. 28, the unity-gain marker gives

$$\boxed{f_{gc,B} \approx 50.6084 \text{ MHz}}. \quad (110)$$

Thus, the marked unity-gain frequency can be used as the gain-crossover frequency of the displayed forward amplifier magnitude.

The phase marker on the same graph is at approximately

$$50.5929 \text{ MHz}, \quad (111)$$

where the plotted phase is approximately

$$90.9667^\circ. \quad (112)$$

The gain-crossover marker itself is at 50.6084 MHz.

However, this value should not be interpreted as a formal feedback-loop phase margin because the plotted quantity is the forward amplifier transfer function rather than a closed-loop feedback transfer function.

Similarly, a numerical gain margin requires a clearly identified phase-crossover frequency at which the loop phase is  $-180^\circ$ .

Therefore, the directly measurable values from the supplied Part-B graph are

$$f_{3dB,B} \approx 951.617 \text{ kHz} \quad (113)$$

and

$$f_{gc,B} \approx 50.6084 \text{ MHz}. \quad (114)$$

The gain margin and phase margin are defined conceptually above but are not assigned numerical values without a loop-gain measurement.

## Section 14: Comparison and Conclusion

**Table 7:** Comparison of Part A and Part B

| Parameter            | Part A: Resistive Load | Part B: Active Load                                       |
|----------------------|------------------------|-----------------------------------------------------------|
| Load type            | Resistor               | PMOS current-mirror / active load                         |
| Specified gain       | 4 V/V                  | 4 V/V                                                     |
| Simulated DC gain    | 3.84258 V/V            | 51.9063 V/V                                               |
| Simulated gain       | 11.69 dB               | 34.30 dB                                                  |
| −3 dB frequency      | 25.3599 MHz            | 951.617 kHz                                               |
| Unity-gain frequency | 95.0691 MHz            | 50.6084 MHz                                               |
| NMOS width           | 8.18 $\mu\text{m}$     | 36.96236 $\mu\text{m}$                                    |
| PMOS width           | –                      | 99.75454 $\mu\text{m}$                                    |
| Main advantage       | Simple structure       | Higher voltage gain                                       |
| Main trade-off       | Lower gain             | Large output resistance and lower dominant-pole frequency |

### 14.1 Part-A Conclusion

The Part-A design target was

$$|A_V| = 4 \text{ V/V}. \quad (115)$$

Using the  $g_m/I_D$  methodology and the MOS characterization data, the calculated transistor width was

$$W_A \approx 8.18 \text{ } \mu\text{m}. \quad (116)$$

The calculated transconductance and drain resistance were

$$g_m = 3.14 \text{ mS} \quad (117)$$

and

$$R_D \approx 1.52 \text{ k}\Omega. \quad (118)$$

The pole-zero simulation gives a DC gain of

$$3.84258 \text{ V/V} \quad (119)$$

---

or

$$11.69 \text{ dB.} \quad (120)$$

Hence, the actual simulated gain is slightly below the target of 4 V/V by approximately 3.94%.

The supplied frequency-response plots show a  $-3$  dB frequency of 25.3599 MHz and a unity-gain frequency of 95.0691 MHz.

## 14.2 Part-B Conclusion

The active-load CS amplifier uses separate NMOS and PMOS width calculations.

The selected widths are

$$W_N \approx 36.96236 \text{ } \mu\text{m} \quad (121)$$

and

$$W_P \approx 99.75454 \text{ } \mu\text{m.} \quad (122)$$

The simulated DC gain is

$$51.9063 \text{ V/V} \quad (123)$$

or approximately

$$34.30 \text{ dB.} \quad (124)$$

Therefore, the active-load amplifier provides a gain substantially higher than 20 dB.

The supplied gain/phase plot gives

$$f_{3\text{dB},B} = 951.617 \text{ kHz} \quad (125)$$

and

$$f_{u,B} = 50.6084 \text{ MHz.} \quad (126)$$

The gain at the marked  $-3$  dB point is approximately 31.3043 dB, and the gain at the unity-gain crossing is 0 dB.

## 14.3 Final Observation

The experiment demonstrates the difference between resistive and active loading in common-source amplifier design.

The resistive-load stage provides a gain close to the specified 4 V/V target, while the active-load stage provides a much larger gain of approximately 34.30 dB.

The  $-3$  dB bandwidth, dominant pole and unity-gain frequency describe the forward amplifier frequency response. Gain margin and phase margin are included as control-system concepts, but formal numerical values require a defined feedback-loop transfer function.